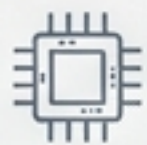
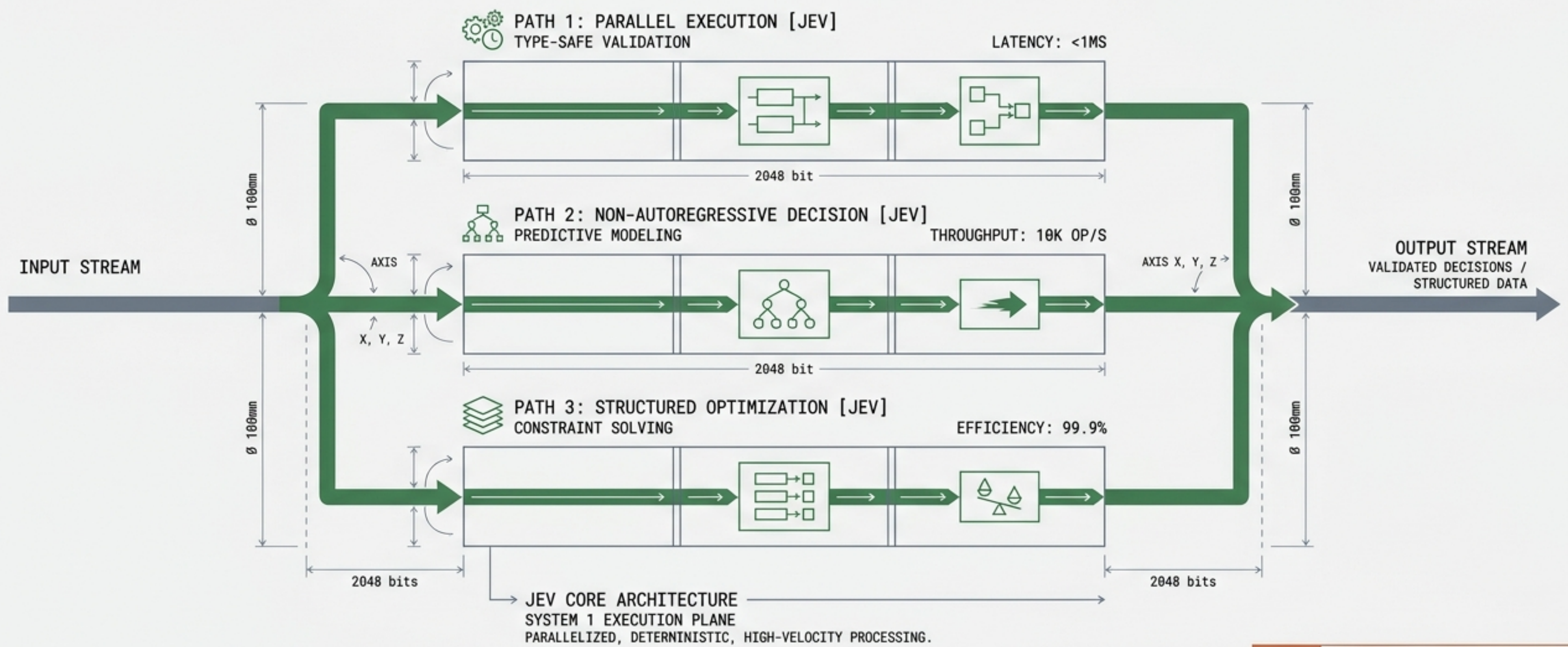


MACHINE-NATIVE INTELLIGENCE

ARCHITECTURAL FOUNDATIONS OF NON-AUTOREGRESSIVE DECISION ENGINES & TYPESAFE JEV



SYSTEM STATUS: ONLINE
PARADIGM: DUAL-PLANE /
SYSTEM 1 EXECUTION

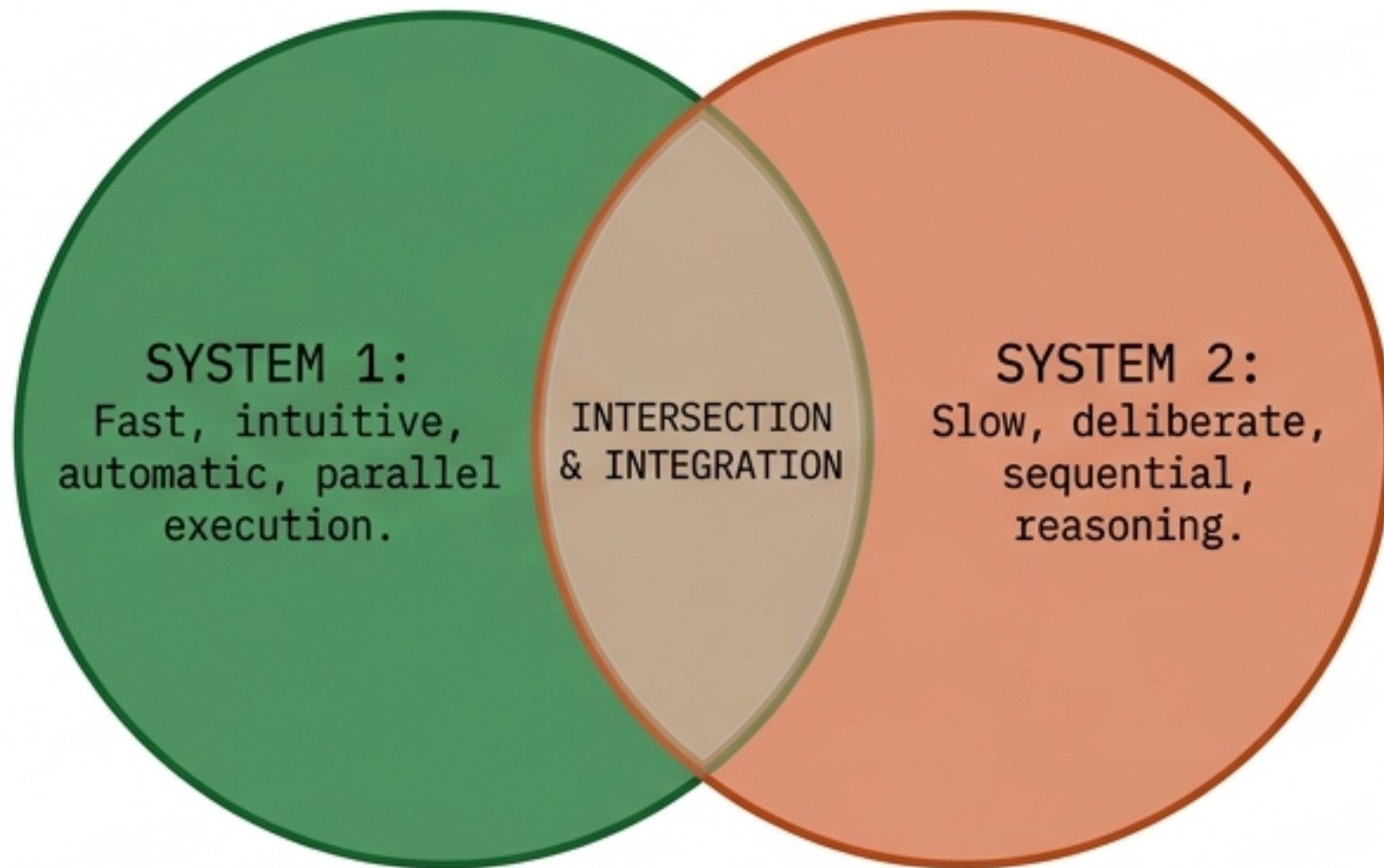




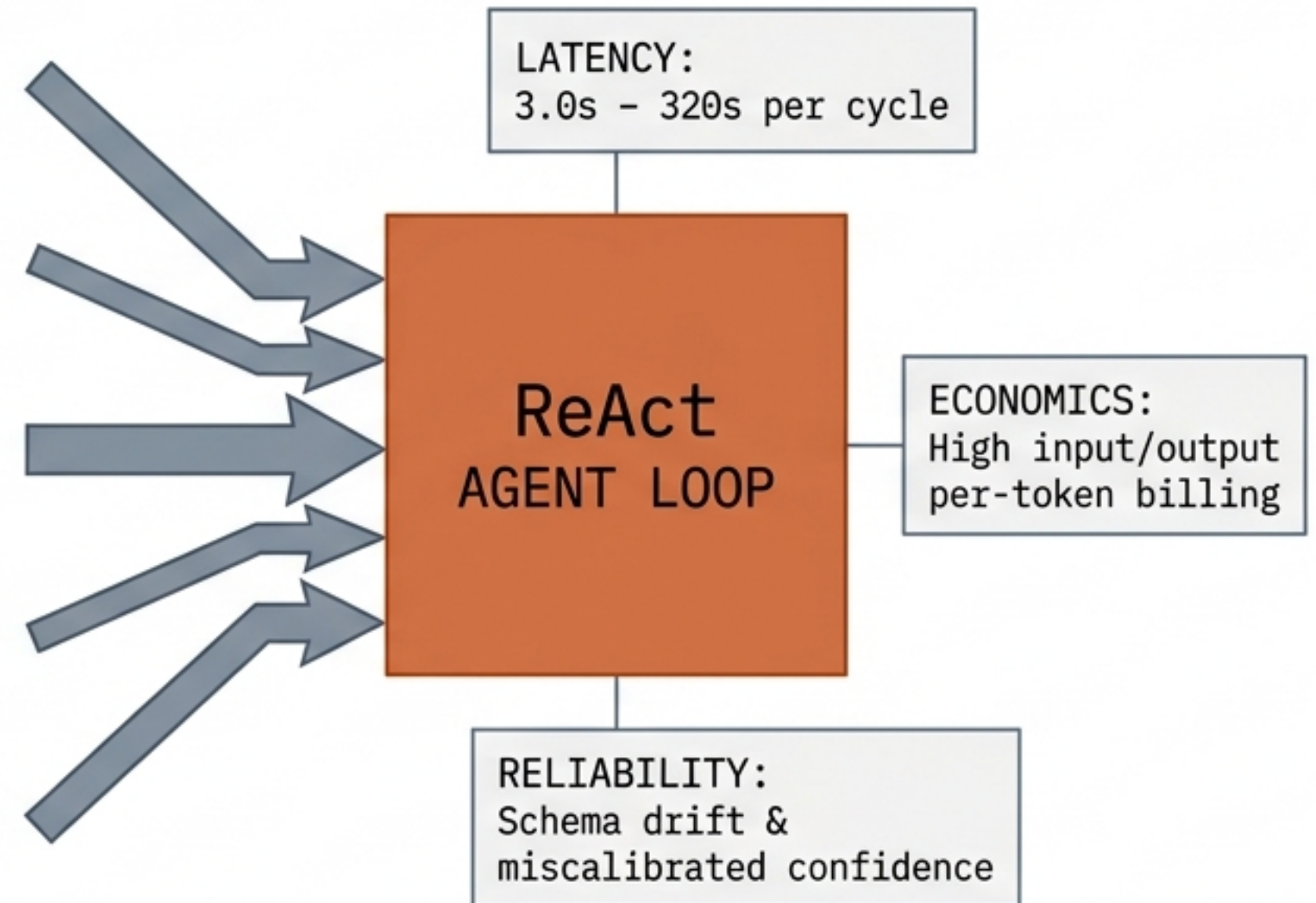
INTEGRATION NOTE
SYSTEM 2 (FRONTIER LLNS) INTERFACE FOR
COMPLEX REASONING (SEQUENTIAL, CONTEXTUAL)
IS SEOREGATED AND ACCESSED VIA DEDICATED
API GATEWAYS.

THE COGNITIVE DIVIDE VS. THE MONOLITHIC BOTTLENECK

THE COGNITIVE BLUEPRINT



THE MONOLITHIC BOTTLENECK



Generative AI excels at open-ended reasoning but introduces massive friction when forced into deterministic software control paths.

THE SYSTEM ONE PARADIGM: TYPESAFE JEV

THE ORIGIN

- **DEVELOPER:** TypeSafe AI (founded by Diogo Almeida, former OpenAI researcher).
- **BACKING:** \$40M seed funding led by DCVC (Sept 2026).
- **NOMENCLATURE:** References the Jevons Paradox—as marginal cost drops, automated decision volume scales exponentially.

THE PARADIGM SHIFT

NO STRING GENERATION

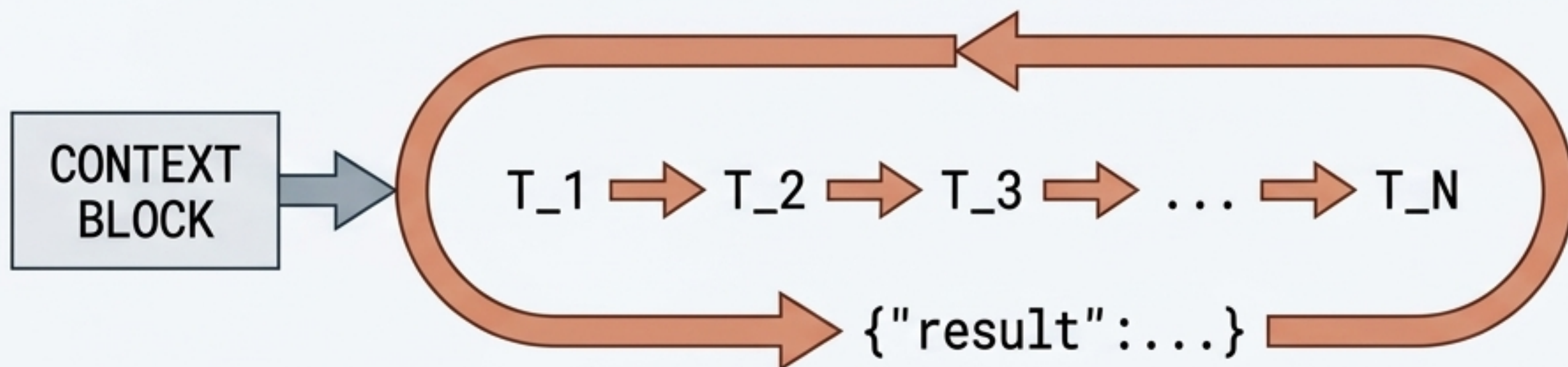
- **MECHANIC:** Ingests unstructured context (up to 64,000 tokens) alongside typed, pre-bounded questions.
- **PROCESSING:** Evaluates all questions simultaneously in a single forward pass.
- **OUTPUT:** Returns mathematically calibrated probabilities and confidence scores, not conversational prose.

THE ENGINE MATRIX: JEV VS. FRONTIER AUTOREGRESSIVE LLMS

	Jev (System 1)	Frontier LLMs
Primary Task	Pure Judgment (Route, triage, gate)	Pure Generation (Synthesis, reasoning, code)
Sampling Mechanic	Parallel Single-Pass	Autoregressive Loop
Latency Profile	70–500ms	3.0s–320.0s+
Economics	\$0.042/MTok (Output Free)	High Input + 3x–5x Output billing
Schema Reliability	100% Guaranteed by Construction	Variable (Regex/JSON-repair required)
Confidence	Calibrated per output	Overconfident / Inconsistent
Hallucination Mode	Selects wrong valid option	Invents invalid facts/syntax

BYPASSING THE DECODER: PARALLEL VS. SEQUENTIAL INFERENCE

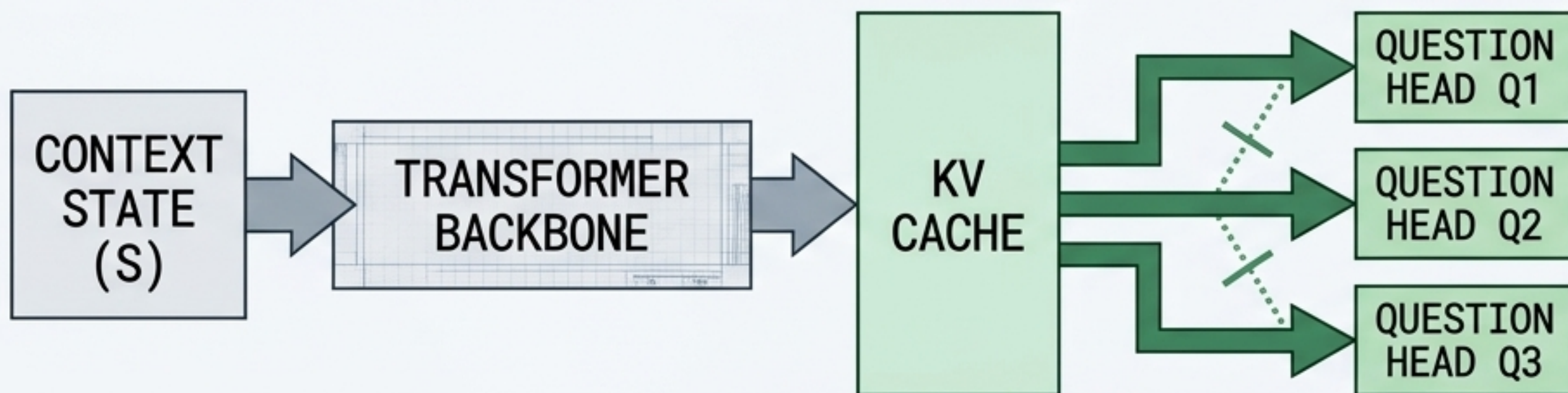
SYSTEM 2: AUTOREGRESSIVE LOOP



Linear generation time.

Repetitive state re-encoding.

SYSTEM 1: NON-AUTOREGRESSIVE SINGLE PASS



$O(S)$ computation.

Zero cross-question contamination.

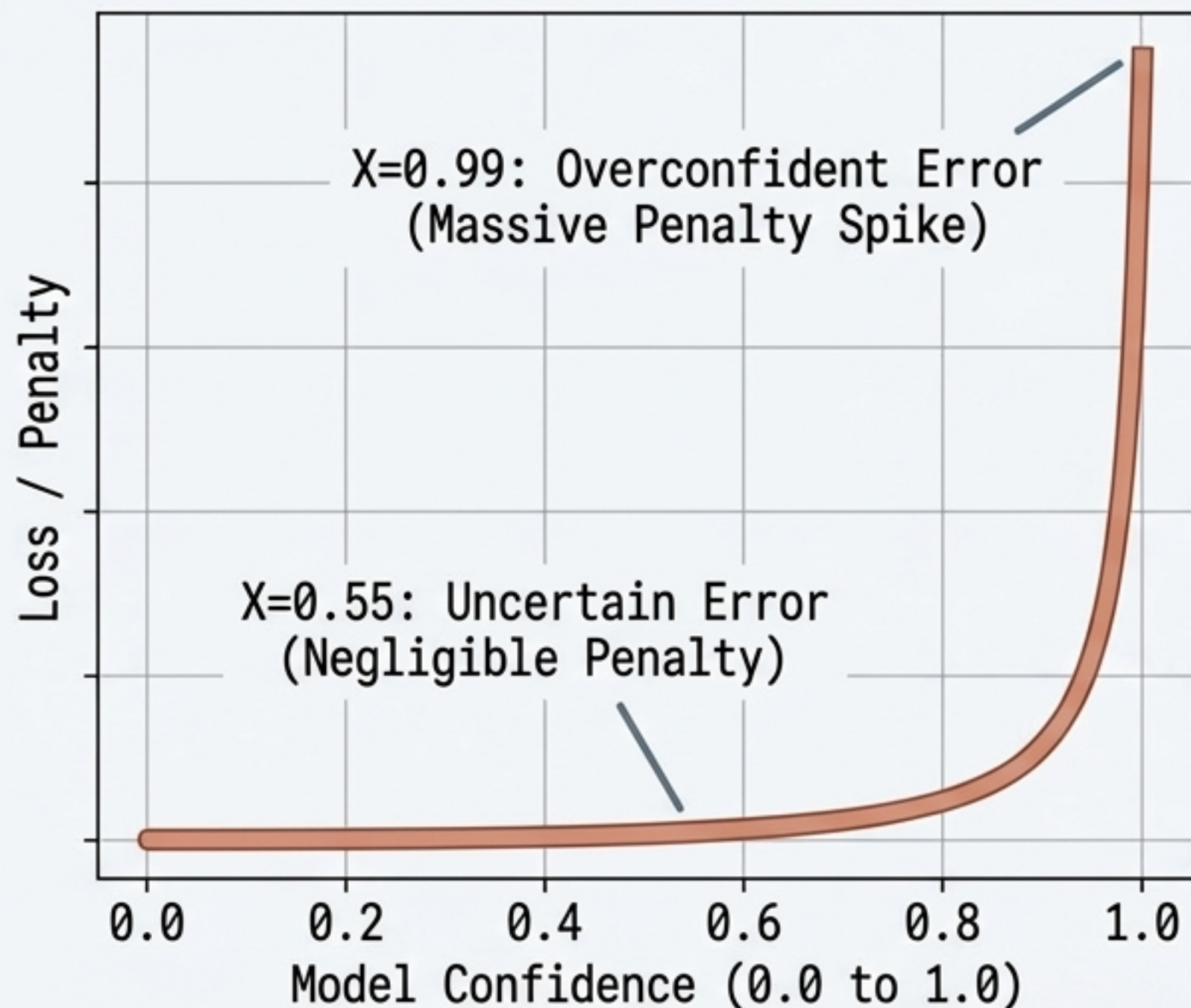
Instant probability readouts.

THE DECISION PRIMITIVES: SCHEMA CONFORMANCE BY CONSTRUCTION

CHOICE	SCORE	NOUL
 Definition: Discrete Categorical Selection. Constraints: 2 to 255 explicit candidate strings. Payload: Selected key, discrete probability distribution ($\Sigma = 1.0$), and aggregate confidence score. Best For: Support ticket routing, intent classification, tool choice selection.	 Definition: Ordered Gradient Rubric. Constraints: 2 to 10 ordered benchmark levels. Payload: Expected numeric value, discrete level distribution, confidence score. Best For: Lead prioritization, sentiment severity, content moderation grading.	 Definition: Binary Proposition Determination. Constraints: Binary boolean state. Payload: Calibrated affirmation scalar $P(\text{yes}) \in [0, 1]$. Best For: Guardrail validation, citation checking, high-risk pre-checks.

MATHEMATICAL CALIBRATION: THE RLCD ASYMMETRIC LOSS LANDSCAPE

THE CALIBRATION CURVE



RLCD MECHANICS

Objective: Epistemic calibration
 $P(\hat{Y}=Y \mid \hat{P}=p) = p.$

Scoring Rule: Discards simple rewards for Strictly Proper Scoring Rules (Cross-Entropy + scaled Brier Score).

Exploration: Injects Gaussian noise directly into logit layers via GRPO.

Post-Hoc: Refines probabilities via L-BFGS temperature scaling.

Abstention Head: Explicit logit slot for out-of-distribution contexts.

OPERATIONAL TELEMETRY: BENCHMARKS & ECONOMICS

10x–100x

Speedup (Latency)

100x–500x

Reduction (Cost)

85%–92%

Absolute Accuracy
Retained

Task	Jev (System 1)	Frontier LLM (System 2)	Result Synthesis
Vendor Eval (Support Triage)	0.4s, \$0.0004	23.3s, \$0.0836	Jev retains 91% of accuracy at 1/200th cost.
Dan Shipper Evals (Defect Check)	0.35s median	8.83s	Cost 580x lower. Jev caught 6/7 defects vs LLM 7/7.
Every Evals (Legal Reranking)	777 docs in <0.70s	45.0s+	Top-10 recall lifted from 38% to 62%.
pi-warden Audit (17k tool calls)	~250ms inline	1.80s+	88% precision on execution holds.

THE JAGGEDNESS PROFILE: BOUNDARIES AND FAILURE MODES

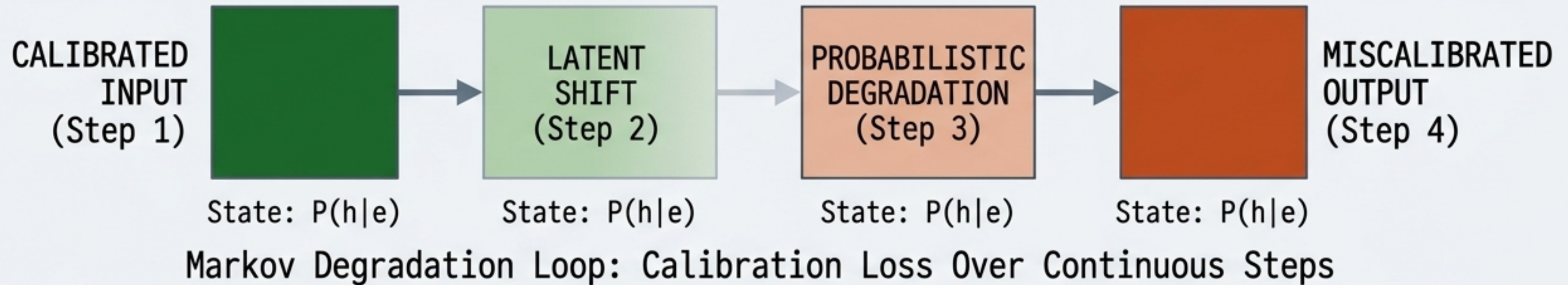
THE SWEET SPOT (GREEN PATHS)

- ✓ Inline programmatic hooks & real-time game loops.
- ✓ High-volume batch mapping and feature extraction.
- ✓ Zero-syntax-error routing and intent classification.
- ✓ Guarding and intercepting LLM tool calls.

THE FAILURE MODES (RED PATHS)

- ⚠ Literal Interpretation: Cannot infer unstated developer intent.
- ⚠ Math & Counting Deficits: Cannot tally items (e.g., 'more than 5 logs').
- ⚠ Temporal Blindness: Parses dates as raw strings, not chronological timestamps.
- ⚠ Multi-Hop Limits: Fails at spatial geometry, long-horizon planning.
- ⚠ Zero Explainability: Outputs numbers, not human-auditable rationales.

THEORETICAL LIMITATION: NON-COMPOSABILITY OF CALIBRATION



THE DOAN NGOC CRITIQUE: “CALIBRATION DOES NOT COMPOSE, TYPES DESTROY VAGUENESS.”

THE PRINCIPLE: Epistemic calibration under RLCD is a localized property holding strictly at a single hop of static data.

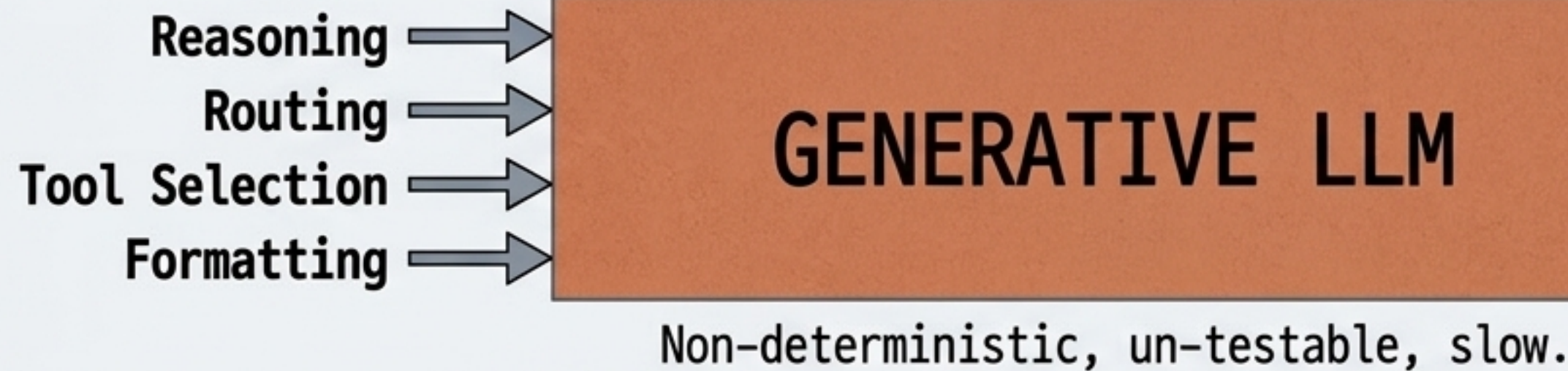
THE MARKOV DEGRADATION: In multi-step continuous agent loops, conditional probabilities degrade due to latent state distribution shifts.

THE ENGINEERING REALITY: Aggregate calibration across loops requires continuous **post-hoc re-calibration**. Primitive probabilities do not compound perfectly over time.

SECURITY ADDENDUM: Probabilistic evaluation does not eliminate the need for deterministic security controls (allowlists, privilege checks).

ARCHITECTURAL SYNTHESIS: SPLIT DECISION AND GENERATION PLANES

THE MONOLITHIC BOTTLENECK



THE DUAL-PLANE ARCHITECTURE

GENERATION PLANE (System 2 / Frontier LLMs)

Invoked infrequently for high-context synthesis, creative writing, multi-hop reasoning, and code.

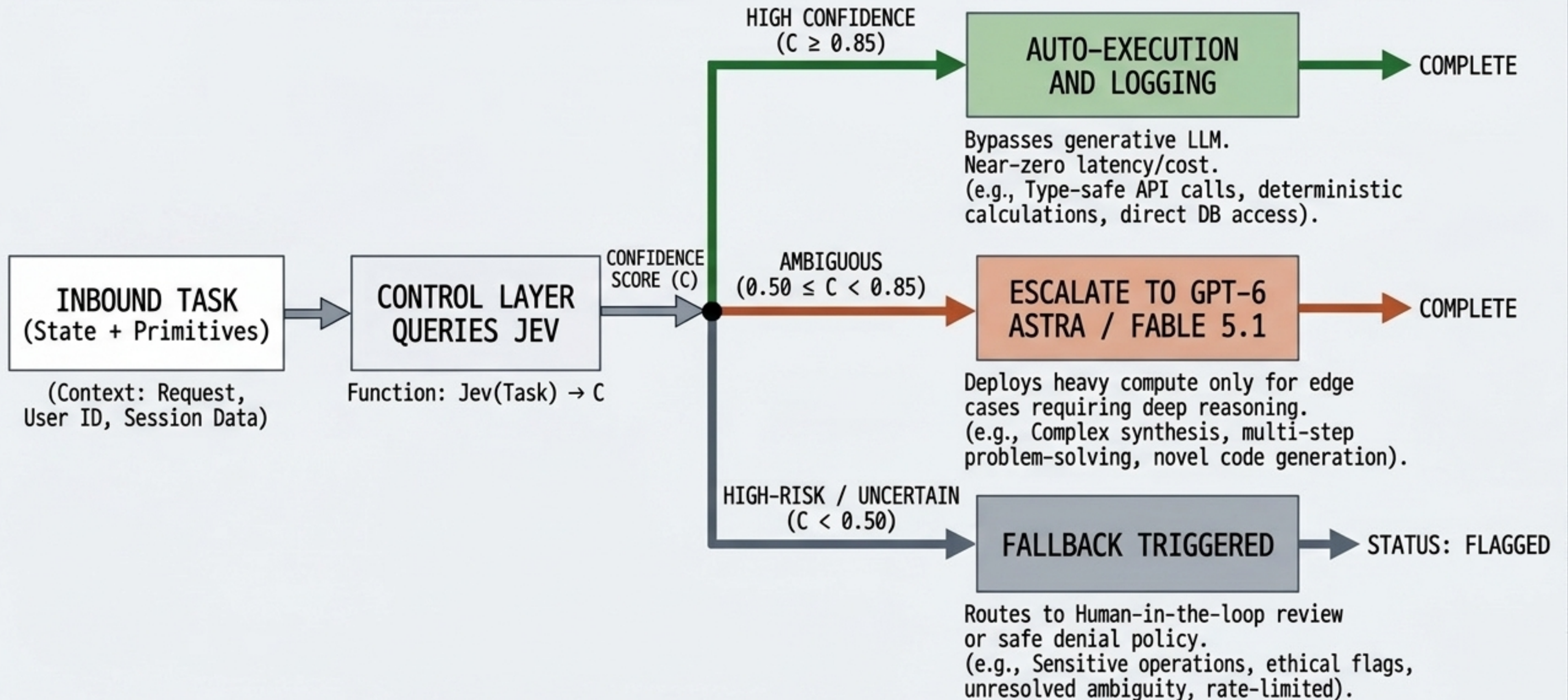
CONTROL LAYER (Python/TypeScript)

The programmatic bridge. Numeric thresholds, retry loops, deterministic safety allowlists. Diffable and unit-testable.

DECISION PLANE (System 1 / Jev)

Handles 80-90% of traffic. Pure judgment: route, triage, score, gate. Explicit execution in sub-seconds.

THE ESCALATION CASCADE: ROUTING BY CALIBRATED CONFIDENCE



THE ECOSYSTEM: FRAMEWORKS, TOOLING, AND REPLICATION

VERCEL AI SDK 7 & GATEWAYS

- Native support via the `'evaluate'` method.
- Allows developers to execute typed primitive queries inline with standard `LLM` calls seamlessly.

OPENJEV (Verdict-open-jev)

- Open-source reverse-engineering built on `ModernBERT-base` (151M parameters).
- Executes in `<35ms` locally; runs client-side in `WebGPU/WASM`.
- Achieves `0.845` modal agreement with `Jev`.

ENTERPRISE INTEGRATIONS

- Spring AI TypeSafe: Java ecosystem wrappers (`'JevJudge'`) for enterprise routing.
- DSPy (`'dspy-typesafeify'`): Programmatic optimization against `RLCD` loss surfaces.
- MCP (`'typesafe-mcp'`): Tool servers for autonomous inline safety checks.

IMPLEMENTATION FRAMEWORK: INTEGRATING THE DECISION PLANE

STEP 1: IDENTIFY BOUNDED DECISION POINTS

Audit existing LLM pipelines for repeated, high-volume classification tasks currently using forced JSON-mode (e.g., ticket triage, tool filtering).

STEP 2: DEFINE RISK-ADJUSTED THRESHOLDS

Map actions on a dual-axis matrix: Consequence Severity vs. Model Confidence.

Establish empirical floors before enabling auto-execution.

STEP 3: DEPLOY THE DUAL-PLANE ARCHITECTURE

Shift simple routing and gating to **Jev**. Reserve expensive **Frontier LLMs** strictly for the generation plane and low-confidence escalations.

STEP 4: RETAIN DETERMINISTIC BACKSTOPS

Never rely purely on probabilistic scoring for critical security.

Maintain hard-coded allowlists, regex interceptors, and relational permissions in code.